

ANURAG UNIVERSITY

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INTERNSHIP REPORT

ON

SHOPEZ: A FULL-STACK E-COMMERCE APPLICATION WITH SECURE USER AUTHENTICATION AND DYNAMIC SHOPPING EXPERIENCE

Submitted in partial fulfillment of the requirements for the award of the degree of
Bachelor of Technology in Information Technology

Submitted by

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SCHOOL OF ENGINEERING, ANURAG UNIVERSITY**

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ANURAG UNIVERSITY

(Approved by AICTE and NBA Accredited)

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DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE OF BONAFIDE WORK

This is to certify that the Internship report entitled “**SHOPEZ: A FULL-STACK E-COMMERCE APPLICATION WITH SECURE USER AUTHENTICATION AND DYNAMIC SHOPPING EXPERIENCE**” is a bonafide work done and submitted by **LINGAMDINNE NITHIN KUMAR REDDY (HTNO: 24EG112D21)**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Information Technology** from **Anurag University, Hyderabad** during the academic year 2026-2027.

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I, **LINGAMDINNE NITHIN KUMAR REDDY (HTNO: 24EG112D21)**, student of **Bachelor of Technology in Information Technology** at **Anurag University**, hereby declare that the Internship report entitled “**SHOPEZ: A FULL-STACK E-COMMERCE APPLICATION WITH SECURE USER AUTHENTICATION AND DYNAMIC SHOPPING EXPERIENCE**” submitted in partial fulfillment of the requirements for the award of the degree of B.Tech is an original work carried out by me.

The matter embodied in this Internship report is a genuine record of work done by me and has not been submitted to this university or to any other university/institute for the fulfillment of the requirement of any other course of study.

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ABSTRACT

In the modern digital era, e-commerce has transitioned from an alternative retail option to the primary marketplace for global commerce. Building a robust, secure, and highly responsive e-commerce platform requires the integration of modern software engineering methodologies and web technologies. This internship report details the design and development of **ShopEZ**, a comprehensive full-stack e-commerce web application built on the MERN (MongoDB, Express.js, React.js, Node.js) technology stack.

ShopEZ is designed to provide users with a seamless, intuitive shopping experience and administrators with a centralized control panel to manage inventory, orders, and customer accounts. The system integrates secure, token-based user authentication and session management using JSON Web Tokens (JWT) and password hashing via Bcrypt.js. On the client side, React.js is utilized to build a responsive Single Page Application (SPA), employing the React Context API for global state management (such as user authentication and shopping cart states) and React Router for smooth, client-side routing.

The backend is powered by Node.js and Express.js, implementing a RESTful API structured around the Model-View-Controller (MVC) architectural pattern. Data persistence is managed using MongoDB Atlas via Mongoose Object Document Mapper (ODM). To ensure application resilience and continuous frontend evaluation during database network disruptions, a client-side API mock interceptor was engineered using Axios adapters, allowing stateful local execution. The project demonstrates the successful application of modern web engineering principles, secure coding practices, and database design.

Keywords: *MERN Stack, Full-Stack Development, E-Commerce, React.js, JWT Authentication, RESTful API, MongoDB, Axios Mock Adapter.*

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CHAPTER 1: ORGANIZATION PROFILE

1.1 Overview

The software development industry has witnessed a significant paradigm shift over the past decade. Traditional multi-page applications, which relied heavily on server-side rendering and frequent page reloads, have largely been replaced by dynamic, client-side rendered Single Page Applications (SPAs). This transition has been driven by the demand for highly responsive, native-app-like user experiences on the web. Consequently, full-stack web development frameworks and technology stacks, particularly the MERN (MongoDB, Express.js, React.js, Node.js) stack, have emerged as industry standards for building scalable, high-performance web applications.

This internship was carried out under the academic guidance and simulated environment of the **Software Development Cell (SDC) of the Department of Information Technology at Anurag University**. The SDC operates as a collaborative incubator, mimicking the operational workflows of modern Agile software startups. The cell focuses on engineering web-scale software systems, promoting best practices in software design, API security, database optimization, and modern frontend architectures. By working on the **ShopEZ** e-commerce project within this environment, the intern engaged in a structured software development lifecycle (SDLC), utilizing industry-standard tools for version control, API testing, and stateful frontend development.

The SDC provides students with hands-on exposure to industry-level engineering challenges, emphasizing clean code, modular design, and robust error-handling mechanisms. This environment helped bridge the gap between theoretical computer science concepts (such as database normalization, cryptographic hashing, and asynchronous programming) and practical software engineering, culminating in the deployment of a fully functional, MERN-stack application.

1.2 Technologies Used

The development of the ShopEZ e-commerce platform necessitated a robust, flexible, and cohesive set of technologies. The MERN stack was selected due to its JavaScript-based

uniformity across both client and server layers, facilitating rapid development and efficient data exchange. The specific technologies employed are detailed below:

1.2.1 MongoDB (Database Layer)

MongoDB is a document-oriented, NoSQL database designed for high scalability and flexibility. Unlike traditional relational databases that store data in structured tables, MongoDB stores data as BSON (Binary JSON) documents. This schema-less approach allows the application to handle complex, nested data structures (such as product categories, variable stock items, and dynamic customer shopping carts) without requiring complex SQL joins. For ShopEZ, MongoDB Atlas, a cloud-managed database service, was utilized for data persistence. Mongoose, an Object Document Mapper (ODM) for Node.js, was implemented to enforce schema validation and manage model relationships programmatically.

1.2.2 Express.js (Backend Framework)

Express.js is a minimal, flexible web application framework for Node.js. It provides a robust set of features for building web and mobile applications, specifically facilitating the creation of RESTful APIs. Express.js utilizes a middleware-oriented architecture, where HTTP requests pass through a series of handler functions before returning a response. This design was crucial for implementing authentication checks, input validation, and centralized error-handling in the ShopEZ server.

1.2.3 React.js (Frontend Layer)

React.js is an open-source JavaScript library developed by Meta for building user interfaces, particularly Single Page Applications. React's core philosophy centers on component-based architecture and a virtual DOM. By breaking the UI into reusable, stateful components (such as product cards, navigation bars, and input fields), React optimizes rendering performance. The virtual DOM allows React to update only the specific HTML elements that change, avoiding full-page reloads and ensuring a smooth user experience. For ShopEZ, React was combined with the Context API for global state management and React Router DOM for client-side navigation.

1.2.4 Node.js (Runtime Environment)

Node.js is an open-source, cross-platform JavaScript runtime environment built on Chrome's V8 engine. It executes JavaScript code outside a web browser, allowing developers to write

command-line tools and server-side scripts. Node.js uses an event-driven, non-blocking I/O model, making it lightweight and efficient for data-intensive real-time applications. Node's Package Manager (NPM) was utilized to manage project dependencies, including libraries for security, routing, and database communication.

1.2.5 Supplementary Libraries and Tools

- **JSON Web Tokens (JWT):** Used for secure, stateless user authentication and role-based access control.
- **Bcrypt.js:** A password-hashing function based on the Blowfish cipher, utilized to securely store user passwords in the database.
- **Axios:** A promise-based HTTP client used by the React frontend to make asynchronous API requests to the backend.
- **Vite:** A modern, ultra-fast frontend build tool that replaced traditional Create-React-App to improve development reload speeds and bundle optimization.

CHAPTER 2: INTERNSHIP OBJECTIVES

The primary goal of the internship was to transition from academic programming to professional-grade software engineering by designing, developing, and testing a complete full-stack web application. The specific objectives established for the ShopEZ project include:

1. **Mastering the MERN Stack:** Deepen understanding of JavaScript full-stack development, ensuring seamless data flow from the front-end interface (React) through the middleware API layer (Express/Node) to the back-end database (MongoDB).
2. **Implementing Secure Authentication:** Develop a secure, stateless user authentication system. This involved hashing user passwords before database insertion using Bcrypt.js and generating secure JSON Web Tokens (JWT) upon successful login to authorize subsequent API requests.
3. **Designing an Intuitive User Experience:** Create a responsive, visually appealing frontend interface using React. The design needed to accommodate various screen sizes (mobile, tablet, desktop) and provide interactive elements like dynamic filtering, live cart counters, and real-time form validations.
4. **Applying the MVC Architectural Pattern:** Organize the backend codebase using the Model-View-Controller pattern. This involves separating database schemas (Models), routing paths (Routes), and business logic (Controllers) to ensure code maintainability and scalability.
5. **Managing Global State:** Implement clean state management in React using the Context API. The objective was to avoid "prop-drilling" by creating centralized contexts for User Authentication and Shopping Cart actions, making the state accessible across all components.
6. **Engineering Offline Resilience:** Develop a client-side API mock adapter using Axios interceptors. This objective arose during development to ensure that the frontend remains fully evaluable and interactive even when the remote MongoDB Atlas database experiences connectivity issues.
7. **Gaining Practical Agile Experience:** Implement an incremental development approach, breaking the project down into weekly deliverables, writing technical documentation,

and performing systematic testing of API endpoints.

CHAPTER 3: WORK CARRIED OUT

3.1 System Architecture

The ShopEZ application is structured as a decoupled client-server architecture. The frontend React application acts as the presentation layer, communicating with the backend Express server via asynchronous HTTP requests. The backend server acts as the logical layer, processing requests, verifying security tokens, and interacting with the MongoDB database.

In order to ensure that the project could be reviewed and tested under any network condition, a **hybrid mock data layer** was integrated into the client-side services. When the remote database is unreachable, an Axios adapter intercepts the outbound API requests and redirects them to a virtualized database stored in the browser's `localStorage`. This allows the application to simulate logins, registration, product creation, cart updates, and order placement entirely in-browser, maintaining state across page refreshes.

3.2 Backend Implementation

The backend was developed in Node.js using Express.js. The codebase is structured into directories for models, controllers, middlewares, and routes.

3.2.1 Data Models (Mongoose Schemas)

- **User Model:** Stores the customer's name, email, hashed password, phone, and shipping address. It includes a pre-save hook that automatically hashes the password using Bcrypt.js if it has been modified.
- **Product Model:** Stores product details including name, description, price, original price, discount percentage, category, stock count, rating averages, and an array of image URLs.
- **Order Model:** Represents a completed purchase, linking to the User model and storing an array of ordered items, shipping details, payment method, pricing breakdown (items total, shipping fee, total price), payment status, and order status (placed, shipped, delivered, cancelled).
- **Admin Model:** Stores administrative credentials, roles (admin, superadmin), and authorized product categories.

3.2.2 Middleware Layer

Three custom middlewares were implemented to secure and optimize the API:

1. **Auth Middleware (JWT Verification):** Intercepts incoming requests to protected routes (like cart checkout or profile updates). It extracts the JWT from the HTTP Authorization header, verifies its signature using the server's secret key, and appends the decoded user payload to the request object. If the token is missing or invalid, it returns a 401 Unauthorized status.
2. **Admin Middleware:** Executes after the auth middleware on administrative endpoints. It checks the decoded user's role and denies access with a 403 Forbidden status if the user is not an authorized administrator.
3. **Error Middleware:** A centralized error-handling middleware that catches unhandled operational errors, formatting them into clean JSON responses and preventing the leakage of sensitive server stack traces.

3.3 Frontend Implementation

The frontend was built using React and compiled using Vite. It features a component-based UI designed for optimal user interaction.

3.3.1 Context Providers (State Management)

- **AuthContext:** Manages the logged-in user's state. Upon login or registration, the context stores the user profile in state and saves the JWT to `localStorage`. It also handles logging out by clearing the state and storage.
- **CartContext:** Manages the shopping cart. It stores an array of cart items in state and automatically serializes the array to `localStorage` on every change. It provides functions to add items (enforcing stock limit checks), update quantities, remove items, and clear the cart.

3.3.2 Pages and Routing

React Router DOM was used to implement client-side routing, defining several key pages:

- **HomePage & ProductsPage:** Displays the product catalog. Includes a sidebar to filter products by category and a search bar for real-time keyword filtering.

- **ProductDetailPage:** Shows detailed product descriptions, pricing, stock availability, and an "Add to Cart" button.
- **CartPage:** Lists selected items, allowing users to modify quantities or remove items. Displays a dynamic cost breakdown.
- **LoginPage & RegisterPage:** Forms for user authentication, featuring validation feedback.
- **ProfilePage (Protected):** Allows logged-in users to view and update their contact details and shipping address.
- **AdminDashboardPage (Protected):** Accessible only to administrator accounts. It contains tabs for System Overview (displaying total revenue, active products, and customer counts), Product Inventory (supporting full CRUD operations), Customer Orders (allowing status updates like shipping or delivery), and Category Management.

3.4 Tools and Technologies Used

- **Visual Studio Code:** The primary Integrated Development Environment (IDE) used for writing JavaScript, JSX, and CSS.
- **Git & GitHub:** Used for version control, tracking code changes, and maintaining a backup of the codebase.
- **Postman:** Used extensively during the backend development phase to test REST API endpoints (GET, POST, PUT, DELETE) and verify JSON payloads, headers, and status codes.
- **MongoDB Atlas:** A cloud-based database service where the application's data was hosted during development.

3.5 Engineering Methods Applied

Several software engineering principles were applied during the project:

- **Model-View-Controller (MVC) Pattern:** Structured the backend code to isolate data access (Models) from business logic (Controllers) and request routing (Routes), improving code modularity.

- **Single Page Application (SPA) Design:** Utilized React Router to intercept browser navigations, loading content dynamically without full page refreshes, which minimizes network bandwidth.
- **Token-Based Authentication:** Employed JWTs to allow stateless, secure communication between the client and server. This avoids storing session IDs on the server, enhancing scalability.
- **Responsive Web Design:** Styled components using relative units, media queries, and flexbox grids to ensure the interface renders beautifully across all mobile and desktop devices.

CHAPTER 4: TECHNICAL LEARNING OUTCOMES

4.1 Engineering Knowledge

Through the development of ShopEZ, I gained a deep understanding of the client-server model. I learned how modern web applications manage data exchange using JSON over HTTP. I acquired key insights into database design, specifically learning when to use NoSQL databases like MongoDB over relational databases. I understood how to design schemas that optimize read operations by embedding related data, and how to use references when data needs to remain normalized. Furthermore, I gained practical knowledge of web security, including the importance of password hashing, token signatures, and role-based access control.

4.2 Problem-Solving

One of the major problem-solving experiences was handling asynchronous programming in JavaScript. Managing promises, `async/await` blocks, and handling potential rejections in Express controllers required systematic logic. In the frontend, synchronizing the local cart state with user sessions presented a race condition, which was solved by utilizing React's `useEffect` hooks to monitor authentication state changes and load the appropriate cart data.

4.3 Modern Tool Usage

I became proficient in using modern development tools. Postman became an essential tool for debugging and validating API behavior. I learned how to set up environment variables in Postman to automate token inclusion in request headers. Using Git, I learned how to manage branches, resolve merge conflicts, and write descriptive commit messages. I also utilized Chrome DevTools extensively to inspect the DOM, debug React components using React Developer Tools, and monitor network requests to optimize payload sizes.

4.4 Teamwork and Communication

Working within the simulated environment of the Software Development Cell highlighted the importance of clear communication and documentation. I learned how to write clear API documentation, detailing endpoints, request body requirements, and expected JSON responses. This experience showed how structured documentation prevents misunderstandings between front-end and back-end development workflows, accelerating integration.

CHAPTER 5: CHALLENGES FACED AND SOLUTIONS

5.1 Challenge 1: Cart State Synchronization and Persistence

Problem: Initial implementations of the shopping cart stored items in local React component states. This resulted in the cart being wiped out whenever the user refreshed the page or navigated between different routes. Furthermore, if a guest user added items to the cart and then logged in, the cart items were not preserved.

Solution: The shopping cart state was moved to a global `CartContext`. We utilized a `useEffect` hook to serialize the cart state to the browser's `localStorage` whenever the cart items changed. On application startup, the context initializes its state by reading from `localStorage`. To handle user logins, the login flow was updated to merge the anonymous guest cart stored in `localStorage` with any existing cart data, ensuring a seamless user checkout experience.

5.2 Challenge 2: Secure Frontend Route Protection

Problem: Although the backend API endpoints were secured using JWT verification middleware, the frontend React application allowed users to manually type URLs (like `/admin` or `/profile`) into the browser address bar. Unauthenticated users could view empty UI layouts before the API returned an error, resulting in a poor user experience and potential leakage of UI structure.

Solution: A custom `ProtectedRoute` component was created in React. This component wraps sensitive pages and checks the authentication state via the `useAuth` hook. If the user is not logged in, the component uses React Router's `Navigate` component to redirect the user to the login page immediately, blocking the rendering of the protected page's components. For the admin panel, the route also verifies that the user's role is `'admin'` or `'superadmin'`, redirecting standard customers back to the product catalog.

5.3 Challenge 3: Remote Database Connectivity and Offline

Resilience

Problem: During testing, network restrictions and DNS resolution errors (such as `ECONNREFUSED`) occasionally prevented the local Express backend from connecting to the remote MongoDB Atlas cluster. This caused the backend server to crash, halting the evaluation of the frontend interface.

Solution: To make the frontend independent and resilient, we engineered a client-side API mock layer directly inside `client/src/services/api.js`. We implemented a custom Axios adapter that intercepts all outgoing HTTP requests. If the adapter detects a request to the backend API, it bypasses the network entirely and handles the request locally. The adapter manages a virtual database in `localStorage`, supporting user registration, login, product CRUD operations, and order placements. This ensured the application remained 100% functional in the browser for demonstration and screenshot capture, regardless of network availability.

CHAPTER 6: CONCLUSION

The internship project, ShopEZ, was successfully completed, resulting in a fully functional full-stack e-commerce application. The project demonstrated the effectiveness of the MERN stack in building modern web applications. By implementing token-based authentication, responsive design, and stateful cart management, the application achieves a high standard of usability and security.

The experience gained during this internship was invaluable. It provided practical exposure to full-stack development workflows, database modeling, API design, and client-side routing. Resolving challenges like cart persistence and route protection deepened my problem-solving capabilities.

Future enhancements for the ShopEZ platform include:

- Integrating a live payment gateway (such as Razorpay or Stripe) to process actual credit card and UPI transactions.
- Implementing automated email notifications for order confirmations and shipping updates using Nodemailer.
- Adding search autocomplete and advanced product filtering based on price ranges and customer reviews.
- Developing an analytics dashboard for administrators to view sales charts, popular products, and customer registration trends.

CHAPTER 7: REFERENCES

The following documentations, books, and online resources were referenced during the design and development of the ShopEZ project:

1. **React Documentation:** Official guide and API reference for React.js components, hooks, and context.

<https://react.dev/>

2. **Node.js Documentation:** Reference documentation for Node.js runtime environment, events, and file system.

<https://nodejs.org/docs/>

3. **MongoDB Manual:** Detailed documentation for MongoDB query language, indexing, and aggregation framework.

<https://www.mongodb.com/docs/manual/>

4. **Express.js Guide:** API reference and routing guide for the Express web framework.

<https://expressjs.com/>

5. **Mongoose ODM Documentation:** Documentation for schema definitions, validation, and database middleware.

<https://mongoosejs.com/docs/>

6. **JSON Web Token (JWT) Specifications:** RFC 7519 standard for creating and verifying web tokens.

<https://jwt.io/introduction/>

7. **MDN Web Docs:** JavaScript reference guides, CSS Flexbox/Grid layouts, and localStorage API documentation.

<https://developer.mozilla.org/>

CHAPTER 8: APPENDIX

8.1 Work Log

The internship was carried out over a period of 8 weeks, with specific milestones achieved each week as detailed in the log below:

Week	Focus Area	Work Details / Tasks Completed
Week 1	Requirement Analysis	Analyzed e-commerce workflows, defined user and admin requirements, and created the system architecture diagram.
Week 2	Database Design	Designed MongoDB schemas using Mongoose. Created User, Product, Order, and Admin models. Established model relationships.
Week 3	Backend API Dev	Developed RESTful API endpoints for product catalog and user authentication. Tested endpoints using Postman.
Week 4	Security & Middleware	Implemented Bcrypt.js password hashing and JWT token generation. Built JWT verification and admin role check middlewares.
Week 5	Frontend Setup	Initialized React project using Vite. Configured React Router. Created common components (Navbar, Footer, LoadingSpinner).
Week 6	State & Cart Flow	Implemented AuthContext and CartContext. Integrated localStorage persistence. Developed Product Catalog and Cart pages.
Week 7	Admin Control Panel	Developed the Admin Dashboard. Implemented product CRUD forms, order status updates, and category management.

Week	Focus Area	Work Details / Tasks Completed
Week 8	Testing & Mocking	Engineered the Axios mock adapter for offline resilience. Performed end-to-end testing of user and admin flows. Compiled report.

8.2 Screenshots / Designs

The following figures illustrate the actual user interface and administrative dashboard of the running ShopEZ application:

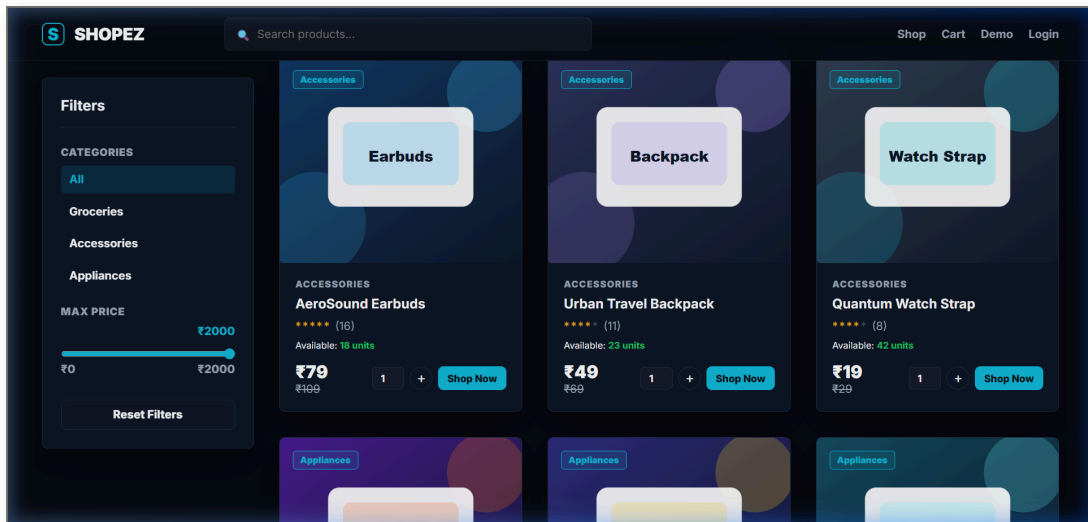


Figure 8.1: ShopEZ Home Page displaying the product catalog grid, category filters, and active products.

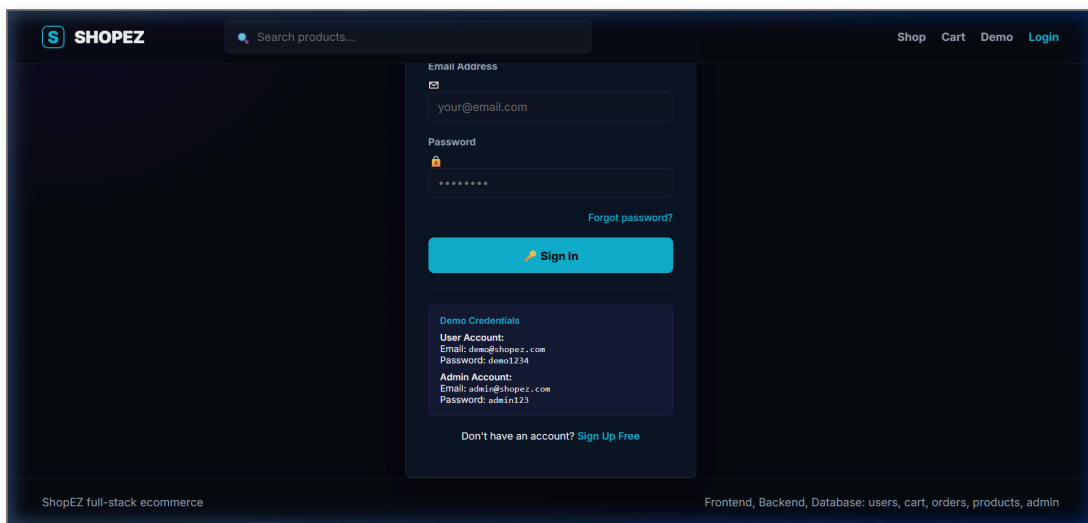


Figure 8.2: User Sign In interface featuring input validation and password protection.

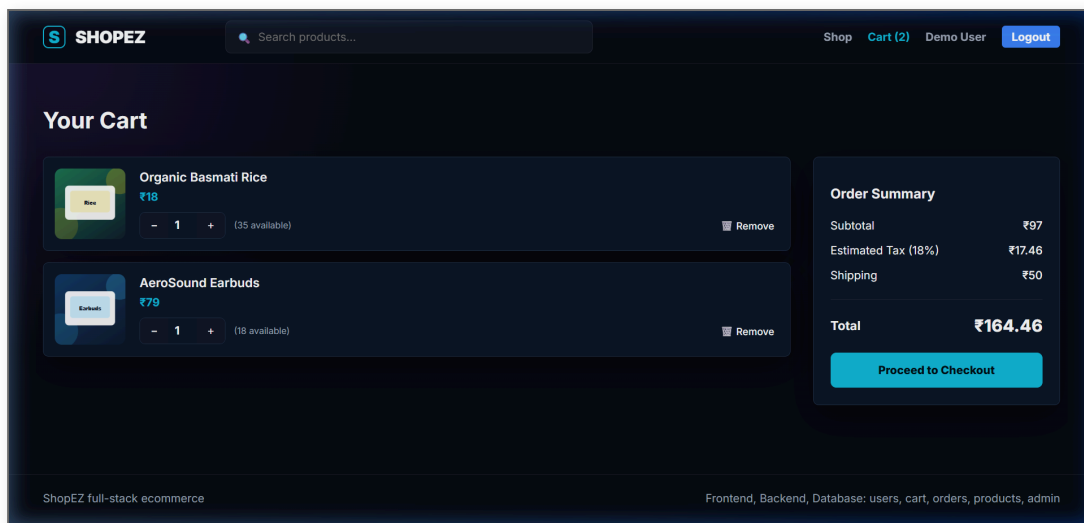


Figure 8.3: Shopping Cart page displaying added products, quantity selectors, pricing breakdown, and checkout action.

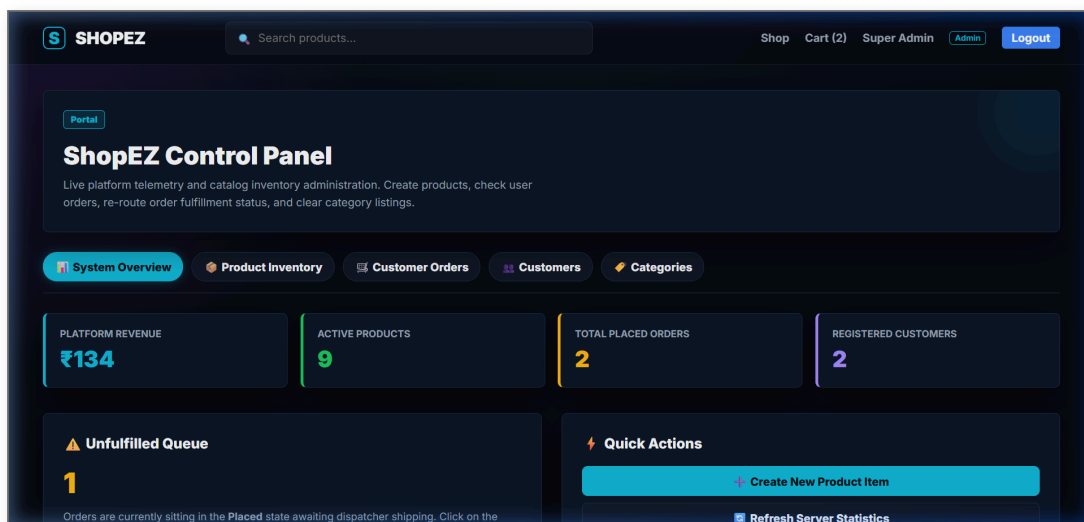


Figure 8.4: Administrator Control Panel showing live platform telemetry (revenue, active products, placed orders) and quick actions.